

QUESTIONS 25 / 25 completed

Remove



REMOVE

Write a C program to remove an element from an array.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[20],n,pos,i;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     scanf("%d",&pos);
9     for(i=pos-1;i<n;i++)
10         a[i]=a[i+1];
11     for(i=0;i<n-1;i++)
12         printf("%d ",a[i]);
13     return 0;
14 }
```

INPUT

```
5
10 20 30 40 50
3
```

QUESTIONS 25 / 25 completed

Insert in Sorted ▾

INSERT IN SORTED

Write a C program to insert an element in a sorted array.

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[20],i,n,x;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     scanf("%d",&x);
9     for(i=n-1;i>0&& a[i]>x;i--)
10         a[i+1]=a[i];
11     a[i+1]=x;
12     for(i=0;i<=n;i++)
13         printf("%d ",a[i]);
14     return 0;
15 }
```

INPUT

5
1 2 3 4 5
6

QUESTIONS 25 / 25 completed

RM Duplicate

RM DUPLICATE

Write a C program to remove all the occurrences of an element from an array.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[20],n,x,i,j;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     scanf("%d",&x);
9     for(i=0;i<n;i++)
10        if(a[i]==x)
11        {
12            for(j=i;j<n-1;j++)
13                a[j]=a[j+1];
14            n--;
15            i--;
16        }
17     for(i=0;i<n;i++)
18         printf("%d ",a[i]);
19     return 0;
20 }
```

INPUT

```
6
1 2 3 2 4 2
2
```

QUESTIONS 25 / 25 completed

Palindrome

PALINDROME

Write a C program to check if an array is a palindrome or not.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[20],i,n,f=1;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     for(i=0;i<n/2;i++)
9         if(a[i]!=a[n-1-i])
10            f=0;
11     if(f)
12         printf("palindrome");
13     else
14         printf("not a palindrome");
15     return 0;
16 }
```

INPUT

5
1 2 3 2 1

QUESTIONS 25 / 25 completed

Common

COMMON

Write a C program to find the common elements between two arrays.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[20],b[20],n,m,i,j;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         scanf("%d",&a[i]);
8     scanf("%d",&m);
9     for(i=0;i<m;i++)
10         scanf("%d",&b[i]);
11     for(i=0;i<n;i++)
12         for(j=0;j<m;j++)
13             if(a[i]==b[j])
14                 printf("%d ",a[i]);
15     return 0;
16 }
```

INPUT

```
5
1 2 3 4 5
4
3 4 6 7
```

QUESTIONS 25 / 25 completed

Add 2 Matrix

ADD 2 MATRIX

Write a C program to add two matrices.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[3][3]={{1,0,0},{0,1,0},{0,0,1}};
5     // int b[3][3]={{-1,3,6},{-4,-6,-8},{-6,-9,-3}};
6     int i,j,flag=1;
7     for(i=0;i<3;i++)
8     {
9         for(j=0;j<3;j++)
10            if(i==j)
11            {
12                if(a[i][j]!=1)
13                    flag=0;
14                break;
15            }
16        else
17            if(a[i][j]!=0)
18            {
19                flag=0;
20                break;
21            }
```

QUESTIONS 25 / 25 completed

Multiply Matrix

MULTIPLY MATRIX

Write a C program to multiply two matrices.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],b[10][10],c[10][10]={0} ,i,j,k,n;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10        for(j=0;j<n;j++)
11            scanf("%d",&b[i][j]);
12    for(i=0;i<n;i++)
13        for(j=0;j<n;j++)
14            for(k=0;k<n;k++)
15                c[i][j]+=a[i][k]*b[k][j];
16    for(i=0;i<n;i++)
17    {
18        for(j=0;j<n;j++)
19            printf("%d ",c[i][j]);
20        printf("\n");
    }
```

INPUT

```
2
1 2
3 4
5 6
7 8
```

QUESTIONS 25 / 25 completed

Transpose

TRANSPOSE

Write a C program to find the transpose of a matrix.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10] ,n,i,j;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10    {
11        for(j=0;j<n;j++)
12            printf("%d ",a[j][i]);
13        printf("\n");
14    }
15    return 0;
16 }
```

INPUT

```
2
1 2
3 4
```


Sum Diagonal

SUM DIAGONAL

Write a C program to find the sum of the diagonal elements of a matrix.

A matrix is a two-dimensional array of numbers arranged in rows and columns. The diagonal elements of a matrix are the elements that lie on the main diagonal, which starts from the

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],n,i,j,sum=0;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10         sum+=a[i][i];
11     printf("%d",sum);
12     return 0;
13 }
```

INPUT

3
1 2 3
4 5 6
7 8 9

QUESTIONS 25 / 25 completed

Product Diag. ▾

PRODUCT DIAG.

Write a C program to find the product of the diagonal elements of a matrix.

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],n,i,j,p=1;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10         p*=a[i][i];
11     printf("%d",p);
12     return 0;
13 }
```

INPUT

```
3
1 2 3
4 5 6
7 8 9
```

QUESTIONS 25 / 25 completed

Sum Up Tri. ▾

SUM UP TRI.

Write a C program to find the sum of the upper triangular elements of a matrix.

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],n,i,j,sum=0;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10         for(j=i;j<n;j++)
11             sum+=a[i][j];
12     printf("%d",sum);
13     return 0;
14 }
```

INPUT

```
3
1 2 3
4 5 6
7 8 9
```

QUESTIONS 25 / 25 completed

Sum Low Tri. ▾

SUM LOW TRI.

Write a C program to find the sum of the lower triangular elements of a matrix.

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],n,i,j,sum=0;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10        for(j=0;j<=i;j++)
11            sum+=a[i][j];
12    printf("%d",sum);
13    return 0;
14 }
```

INPUT

```
3
1 2 3
4 5 6
7 8 9
```

QUESTIONS 25 / 25 completed

Sum Left Diag. ▾

SUM LEFT DIAG.

Write a C program to find the sum of the elements in the left diagonal of a matrix.

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],n,i,j,sum=0;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10         sum+=a[i][n-1-i];
11     printf("%d",sum);
12     return 0;
13 }
```

INPUT

```
3
1 2 3
4 5 6
7 8 9
```

QUESTIONS 25 / 25 completed

Sum Right Diag. ▾

SUM RIGHT DIAG.

Write a C program to find the sum of the elements in the right diagonal of a matrix.

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],n,i,j,sum=0;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10         sum+=a[i][i];
11     printf("%d",sum);
12     return 0;
13 }
```

INPUT

```
3
1 2 3
4 5 6
7 8 9
```

QUESTIONS 25 / 25 completed

Symmetric

SYMMETRIC

Write a C program to check if a matrix is symmetric or not.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],n,i,j,flag=1;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10        for(j=0;j<n;j++)
11            if(a[i][j]!=a[j][i])
12            {
13                flag=0;
14                break;
15            }
16     if(flag)
17         printf("symmetric");
18     else
19         printf("not symmetric");
20     return 0;
21 }
```

INPUT

```
3
1 2 3
2 4 5
3 5 6
```

QUESTIONS 25 / 25 completed

Diagonal or Not ▾

DIAGONAL OR NOT

Write a C program to check if a matrix is diagonal or not.

PROGRAM EDITOR

C ▾

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],n,i,j,flag=1;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10        for(j=0;j<n;j++)
11            if(i!=j&& a[i][j]!=0)
12            {
13                flag=0;
14                break;
15            }
16     if(flag)
17         printf("diagonal");
18     else
19         printf("not a diagonal");
20     return 0;
21 }
```

INPUT

```
3
1 0 0
0 2 0
0 0 4
```


QUESTIONS 25 / 25 completed

Up Tri. Not

UP TRI. NOT

Write a C program to check if a matrix is upper triangular or not.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],n,i,j,flag=1;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10        for(j=0;j<i;j++)
11            if(a[i][j]!=0)
12            {
13                flag=0;
14                break;
15            }
16     if(flag)
17         printf("upper triangular matrix");
18     else
19         printf("not an upper triangular matrix");
20     return 0;
21 }
```

INPUT

```
3
1 2 3
0 4 5
0 0 6
```

QUESTIONS 25 / 25 completed

Low Tri. Not

LOW TRI. NOT

Write a C program to check if a matrix is lower triangular or not.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],i,j,n,flag=1;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10         for(j=i+1;j<n;j++)
11             if(a[i][j]!=0)
12             {
13                 flag=0;
14                 break;
15             }
16     if(flag)
17         printf("lower triangular matrix");
18     else
19         printf("not a lower triangular matrix");
20     return 0;
21 }
```

INPUT

```
3
1 0 0
2 4 0
3 5 6
```

QUESTIONS 25 / 25 completed

Equal

EQUAL

Write a C program to check if two matrices are equal or not.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],b[10][10],n,i,j,flag=1;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10        for(j=0;j<n;j++)
11            scanf("%d",&b[i][j]);
12    for(i=0;i<n;i++)
13        for(j=0;j<n;j++)
14            if(a[i][j]!=b[i][j])
15            {
16                flag=0;
17                break;
18            }
19    if(flag)
20        printf("matrices are equal");
21    else
```

INPUT

```
2
1 2
3 4
1 2
3 4
```

QUESTIONS 25 / 25 completed

Subtract

SUBTRACT

Write a C program to subtract two matrices.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],b[10][10],c[10][10],n,i,j;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<n;i++)
10        for(j=0;j<n;j++)
11            scanf("%d",&b[i][j]);
12    for(i=0;i<n;i++)
13        for(j=0;j<n;j++)
14            c[i][j]=a[i][j]-b[i][j];
15    for(i=0;i<n;i++)
16    {
17        for(j=0;j<n;j++)
18            printf("%d ",c[i][j]);
19        printf("\n");
20    }
21    return 0;
```

INPUT

```
2
5 6
7 8
1 2
3 4
```

QUESTIONS 25 / 25 completed

Multiply Scalar

MULTIPLY SCALAR

Write a C program to multiply a matrix by a scalar.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],i,j,scalar,n;
5     scanf("%d",&n);
6     for(i=0;i<n;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     scanf("%d",&scalar);
10    for(i=0;i<n;i++)
11        for(j=0;j<n;j++)
12            a[i][j]=a[i][j]*scalar;
13    for(i=0;i<n;i++)
14    {
15        for(j=0;j<n;j++)
16            printf("%d ",a[i][j]);
17        printf("\n");
18    }
19    return 0;
20 }
```

INPUT

```
2
1 2
3 4
5
```

QUESTIONS 25 / 25 completed

Determinant

DETERMINANT

Write a C program to find the determinant of a 2x2 matrix.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[2][2],det,i,j;
5     for(i=0;i<2;i++)
6         for(j=0;j<2;j++)
7             scanf("%d",&a[i][j]);
8     det=(a[0][0]*a[1][1])-(a[0][1]*a[1][0]);
9     printf("%d",det);
10    return 0;
11 }
```

INPUT

1 2
3 4

QUESTIONS 25 / 25 completed

Inverse

INVERSE

Write a C program to find the inverse of a 2x2 matrix.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     float a,b,c,d,det;
5     scanf("%f%f%f%f",&a,&b,&c,&d);
6     det=(a*d)-(b*c);
7     if(det==0)
8         printf("inverse doesnot exist");
9     else
10        printf("inverse matrix\n");
11    printf("%.2f %.2f\n",d/det,-b/det);
12    printf("%.2f %.2f\n",-c/det,a/det);
13    return 0;
14 }
```

INPUT

1 2
3 4

QUESTIONS 25 / 25 completed

Rank

RANK

Write a C program to find the rank of a matrix.

PROGRAM EDITOR

C



Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10][10],m,n,i,j,rnk=0;
5     scanf("%d %d",&m,&n);
6     for(i=0;i<m;i++)
7         for(j=0;j<n;j++)
8             scanf("%d",&a[i][j]);
9     for(i=0;i<m;i++)
10    {
11        for(j=0;j<n;j++)
12        {
13            if(a[i][j]!=0)
14            {
15                rn timer++;
16                break;
17            }
18        }
19    }
20    printf("%d",rn timer);
21    return 0;
```

INPUT

```
3 3
1 2 3
0 0 0
2 4 6
```


QUESTIONS 25 / 25 completed

Gauss-Jordan

GAUSS-JORDAN

Write a C program to perform Gauss-Jordan elimination on a matrix.

PROGRAM EDITOR

C

Run

Save

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,j,k,n;
5     float a[10][20],ratio;
6     scanf("%d",&n);
7     for(i=0;i<n;i++)
8         for(j=0;j<n+1;j++)
9             scanf("%f",&a[i][j]);
10    for(i=0;i<n;i++)
11    {
12        for(j=0;j<n;j++)
13        {
14            if(i!=j)
15            {
16                ratio=a[j][i]/a[i][i];
17                for(k=0;k<n+1;k++)
18                    a[j][k]-=ratio*a[i][k];
19            }
20        }
21    }
```

INPUT

```
3
2 1 -1 8
-3 -1 2 -11
-2 1 2 -3
```